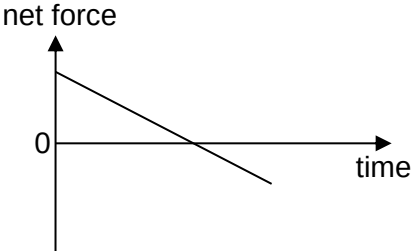
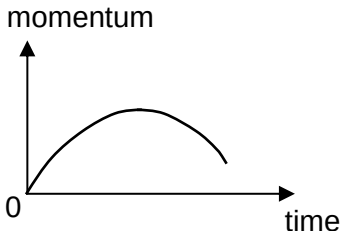
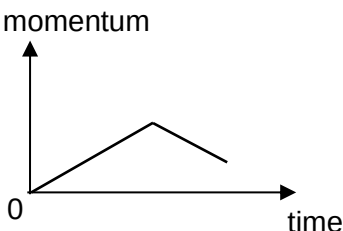
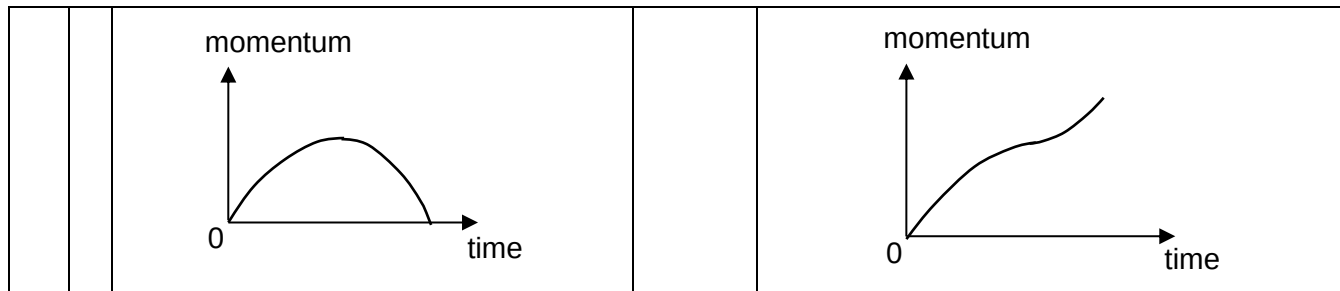


5	A vehicle starts from rest and a net force acts on it. The figure below shows how the net force varies with time.	
		
	Which graph shows how the momentum of the vehicle varies with time?	
	A	B
		
	C	D



Answer: A

$F = dp/dt = \text{gradient of } p\text{-}t \text{ graph}$

As force increases linearly, gradient of $p\text{-}t$ graph increases. As force decreases linearly, gradient of $p\text{-}t$ graph decreases B is incorrect

When F is negative, the momentum will decrease. D is incorrect

Area under $F\text{-}t$ graph is the change in momentum. Hence the increase in momentum in part 1 is greater than that decrease in part 2 B is incorrect